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Bayesian Inverse Transition Learning for Offline Settings

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Summary

- Learning a model is hard under low-coverage data
- Having a model help be more data efficient
- Inverse Transition Learning infers the dynamics using near-optimal demonstrations
- We create a set of constraints to estimate in a gradient-free way an empirical posterior distribution over the transition dynamics T, where each sample has near-optimal guarantees
- Show that planned policies are highly performant and exhibit lower variance across different batches of datas compared to MLE estimates
- Explain how quantifying the uncertainty of the dynamics help planning more informative policies than the expert policy and the one obtained from MLE estimates

What is Inverse Transition Learning?

Infer the dynamics T given near-optimal demonstrations $\mathcal{D}$, the reward function R and a near-optimal expert policy π_ϵ, where π_ϵ is assumed to only take actions that are ϵ-close to the best unkown action a^*.

Why is it hard?

- Number of parameters grows with the dimensions of state and action spaces
- Trade-off between a model that is expressive enough and one that is small enough to learn
- Lack of coverage in offline inverse settings

Why do we care?

- Data efficiency
- Simulation of new data
- Off-Policy Evaluation
- Counterfactual Reasoning

Baselines

- T^{MLE} Maximum Likelihood Estimate obtained from the expert demonstrations $\mathcal{D}$
- $P(T(s'|s,a)|\mathcal{D}) = Dir(N_{s,a} + \mathbf{1}|s,a)$ Probabilistic model over T without making any optimality assumption. Where $\mathbf{1}$ is a vector of 1's of dimension $|S|$ and $N_{s,a}$ is the number of transitions from state s and action a in the batch data $\mathcal{D}$

Constraints

Using the reward R and near-optimal expert policy π_ϵ we create a set of constraints with respect to the dynamics T to impose the values of the actions of the expert to be superior to the ones of unseen actions.

Method

- Sample from $P(T|\mathcal{D})$ and only keep samples that satisfies the constraints.
- Create empirical posterior $P(T|\mathcal{D}, \pi_\epsilon)$ with accepted samples. This will be the estimate of the posterior on T given that the demonstrations from the expert are near-optimal

Metric

$$Q^*_{metric}(\hat{\pi}) = \sum_{s \in S} \left(Q^*(s, a^*) - \sum_{a \in A} \hat{\pi}(a|s) Q^*(s,a) \right)$$

Results

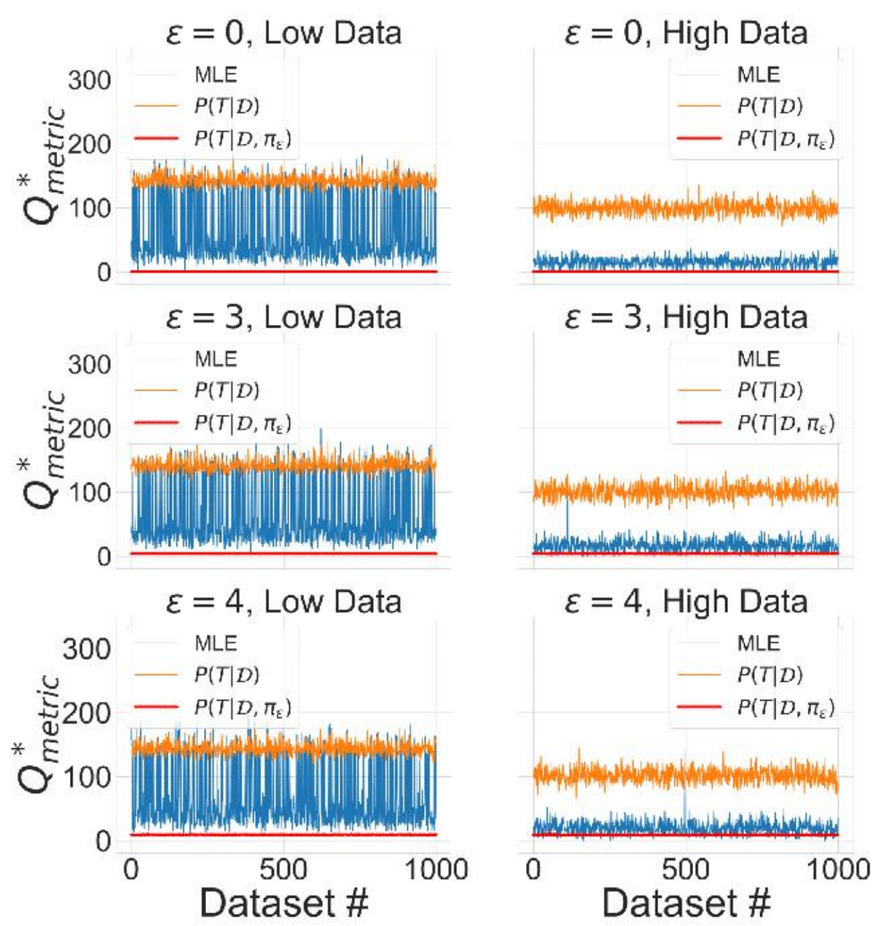
- **MLE**

$$\hat{T} = T^{MLE} \xrightarrow[\text{Value Iteration}]{} (\pi^{MLE}, Q^{MLE}) = (\hat{\pi}, \widehat{Q}) \xrightarrow[\text{Compute Results}]{}$$

- $P(T|\mathcal{D}, \pi_\epsilon)$ **and** $P(T|\mathcal{D})$

$$\{\hat{T}^{(i)}\}_{i=1}^{1000} \sim \text{Posterior} \xrightarrow[\text{Value Iteration}]{} \{(\hat{\pi}^{(i)}, \widehat{Q}^{(i)})\}_{i=1}^{1000}$$

$$\xrightarrow[\text{Empirical Mean}]{} \{(\hat{\pi}, \widehat{Q})\} \xrightarrow[\text{Compute Results}]{}$$



- $P(T|\mathcal{D}, \pi_\epsilon)$ outerperforms both baselines by being more accurate and avoiding low value actions
- $P(T|\mathcal{D}, \pi_\epsilon)$ exhibits significantly less variance over different datasets
- $P(T|\mathcal{D}, \pi_\epsilon)$ allows quantifying uncertainty on corresponding policies

Acknowledgements

This material is based upon work supported by the National Science Foundation under Grant No. IIS-2007076. Any opinions, findings, and conclusions or recommendations expressed in this material are those of the author(s) and do not necessarily reflect the views of the National Science Foundation.